Dear Hiring Manager,

I am a Mechanical Engineering graduate from Clemson University, having completed my Bachelor of Science in Mechanical Engineering in December 2025. I am currently pursuing a Master of Science in Mechanical Engineering through Clemson's BS/MS program, with a specialization in Thermal and Fluid Sciences and a projected graduation in December 2026. I am seeking engineering roles focused on propulsion systems, thermal-fluid analysis, or aerodynamic performance, where I can apply both advanced coursework and hands-on aerospace industry experience.

Through my academic work, I have developed a strong foundation in propulsion, computational fluid dynamics (CFD), compressible flow, and heat transfer. My recent projects include the optimization of a turbojet engine cycle using Brayton cycle analysis, compressible flow relations, and MATLAB-based iterative modeling to determine optimal compressor ratios and afterburner performance. I have also developed CFD-based solvers to analyze compressible nozzle flow, strengthening my understanding of numerical methods and flow physics. Additional work in composite layup optimization using laminate theory and Tsai-Wu failure criteria, along with aerodynamic analysis using XFOIL, has further strengthened my ability to translate theory into quantitative engineering results.

I have also completed three cooperative education rotations at Gulfstream Aerospace Corporation, gaining experience in a highly regulated aerospace environment. My roles spanned vendor project engineering, stress analysis, and baseline design for the G700 program. During these rotations, I developed DO-160 compliant test plans, produced airworthiness documentation, performed structural analyses under FAA Part 25 regulations, and supported design efforts using CATIA and AutoCAD. These experiences strengthened my technical rigor, attention to certification requirements, and ability to communicate engineering results within multidisciplinary teams.

I am motivated to contribute to technically challenging aerospace programs and to continue developing expertise in propulsion, thermal-fluid systems, and aerodynamic performance. I would welcome the opportunity to discuss how my academic focus and industry experience could support your team's engineering objectives.

Thank you for your time and consideration.

Sincerely,
John Vandermeulen